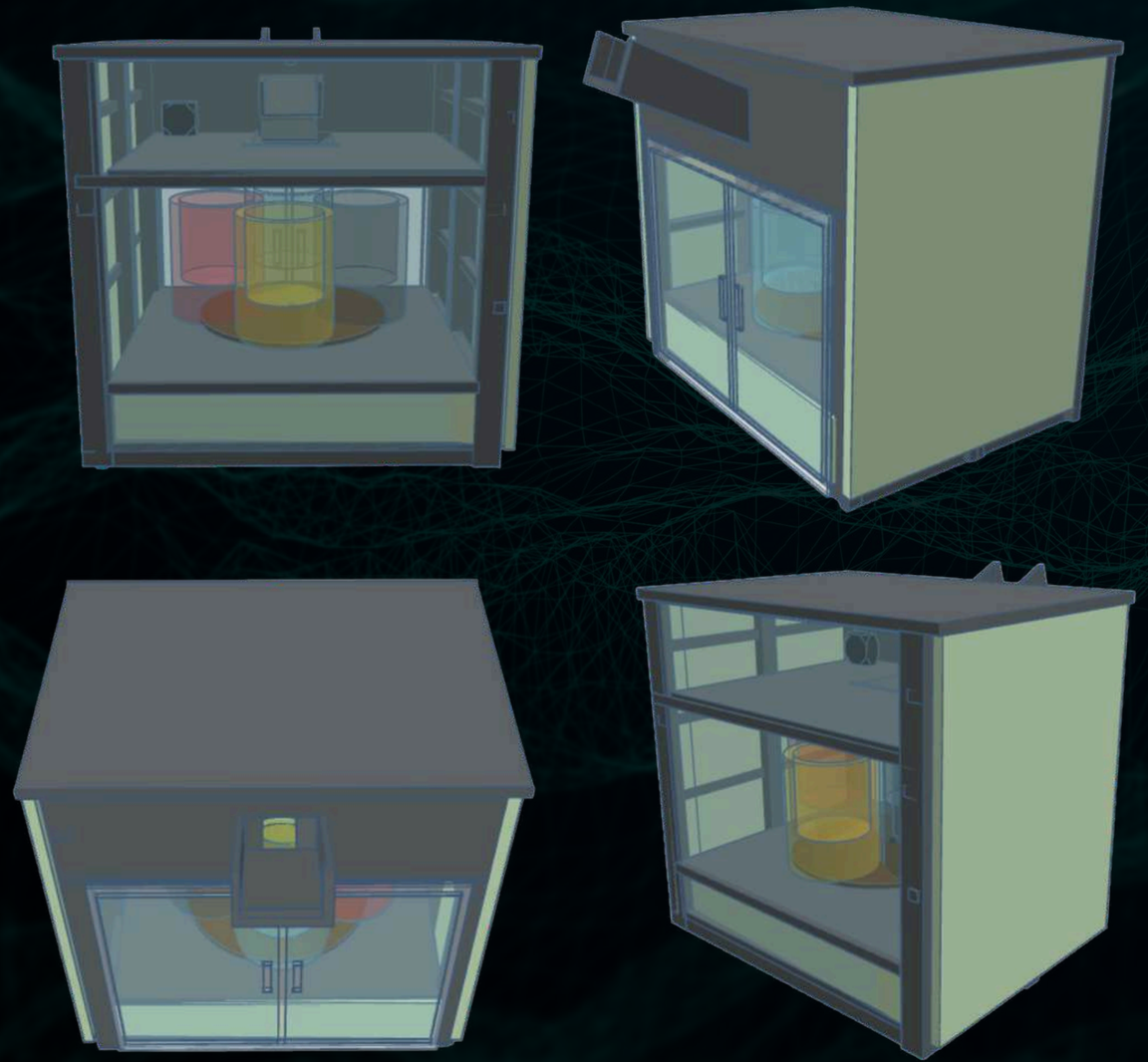
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BINNSIE

Supporting document #2

How the Smart Bin Works



- Binnsie operates as an integrated system that combines sensors, computer vision, and mechanical components to automate waste sorting while creating an interactive learning experience. The bin features a strategically transparent design that allows students to observe the internal process, simulating a live learning lab where technology and sustainability are visibly connected.

- At the top section of the bin, a ramp and an initial gate receive the waste deposited by the user. Once the item enters the system, a camera captures an image, and the AI model identifies the type of material. Based on this classification, a motor located in the lower section rotates to align the waste with the correct internal compartment.

- After positioning is completed, a gate opens, allowing the material to fall by gravity into the appropriate container. This process is repeated continuously, enabling efficient and automated sorting while making each step visible to the user.

- The design also incorporates practical and scalable features. The front and back panels are transparent, to make BINNSIE a live STEAM lab and a learning experience, with a front-facing door that allows easy removal of collected waste once compartments are full. Additionally, the sides of the bin include designated spaces for educational or advertising banners, enabling partnerships and awareness campaigns within the same structure.

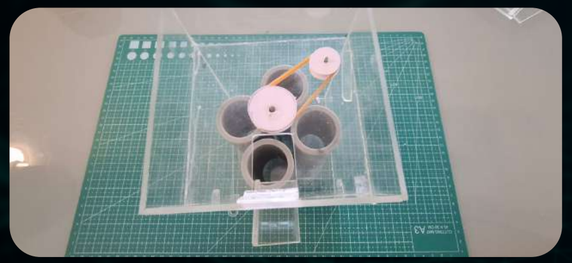
- Through this combination of automation, visibility, and intentional design, Binnsie transforms the act of recycling into both an efficient process and an engaging educational experience.

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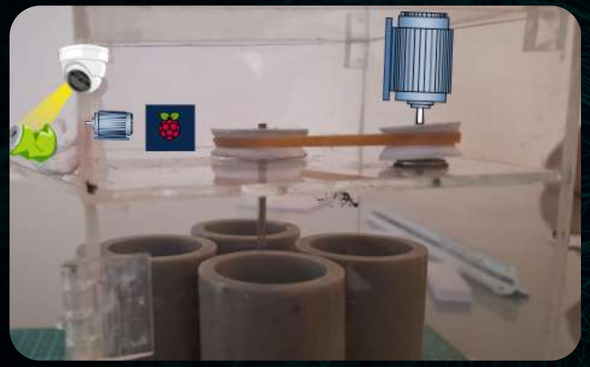
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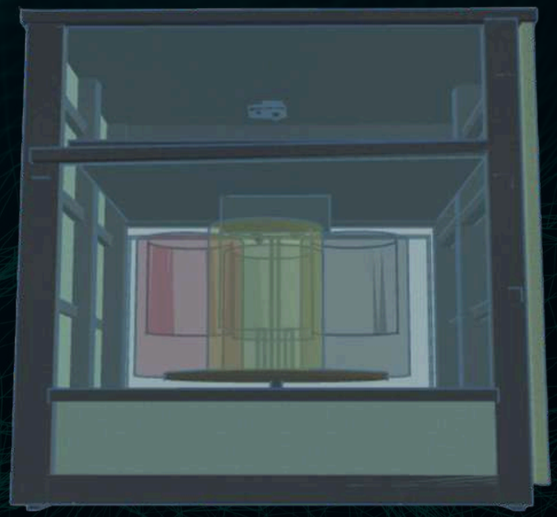
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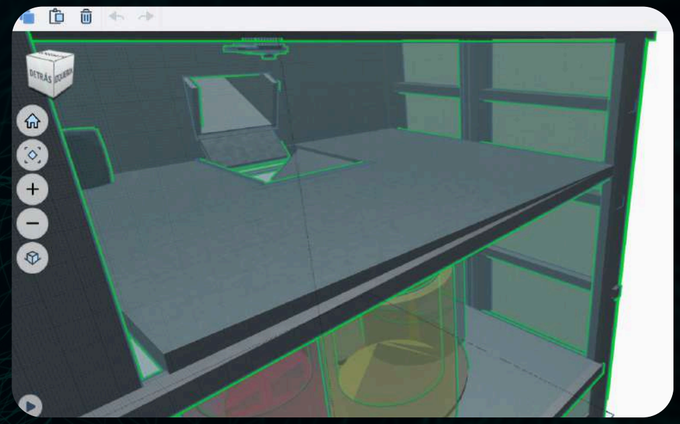
Development Process & Early Prototyping











- The development of Binnsie began with rapid prototyping to test the feasibility of the concept. An initial small-scale transparent prototype made with plastic materials allowed the team to explore how waste moves through the system and identify key challenges such as ramp positioning and release timing. It was at this stage that the rotation mechanism for the four upper bins was first conceptualized and tested at a basic level.

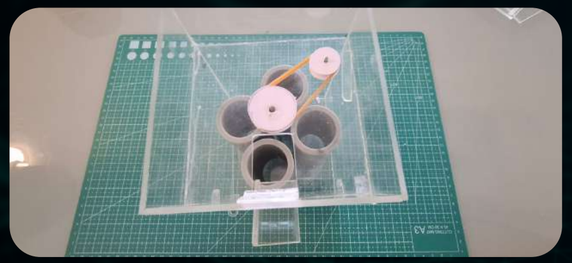
- To further refine the design, a 3D model was developed. This model improved visualization and made it easier to communicate the idea with others and gather feedback. While the motor itself was not fully represented, the model helped guide important design decisions.
- As a result of this process, the motor placement was redesigned from the upper section to the lower part of the bin, and the camera was placed inside the structure. Additionally, the lower section was partially enclosed to protect the mechanism and ensure more reliable operation without compromising the system's overall transparency.
- Always keeping transparent areas for the live learning experience for students.

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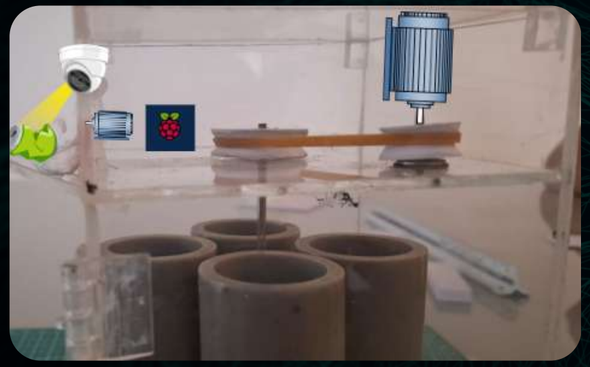
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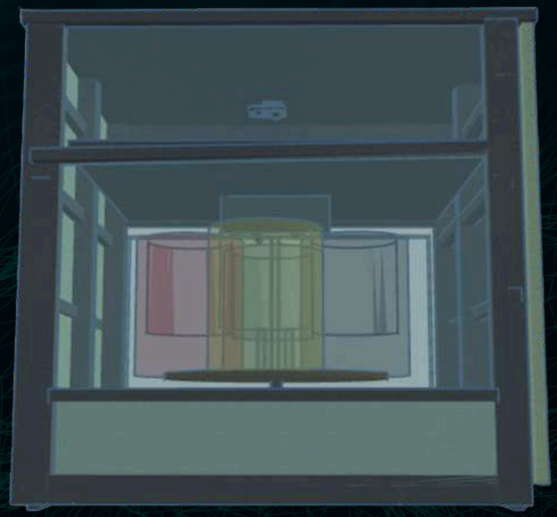
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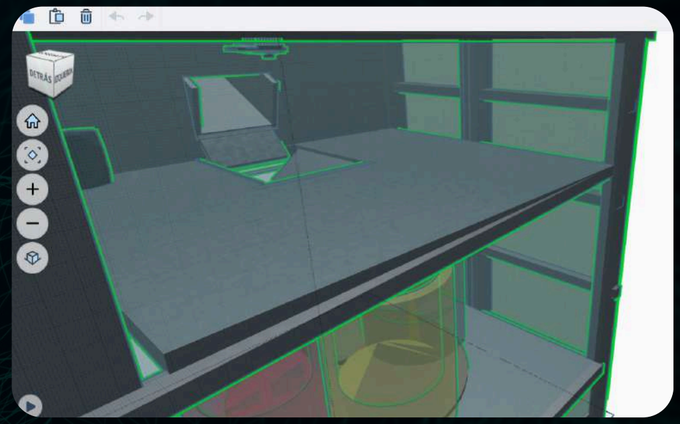
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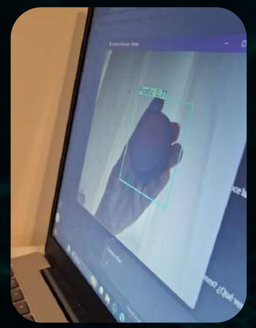
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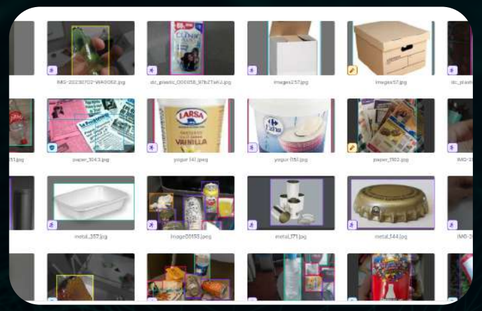
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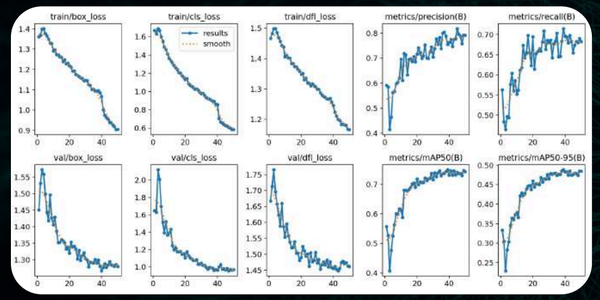
BINNSIE

Supporting document #2

Development Process















- Binnsie uses a computer vision model trained by the team to classify waste in real time. The model was developed using Roboflow for labeling and Google Colab for training, and is deployed on a Raspberry Pi.
- To improve accuracy, we combined public datasets with locally collected data. For this, we placed a cardboard bin in a school for one week to observe and identify the most common types of waste generated by students. This allowed us to understand real consumption patterns and include more relevant materials in the training process.
- The current model achieves a mAP of 0.745, and we aim to reach 0.90–0.95 through further training and dataset expansion.

- Building on multiple iterations and design improvements, we are currently developing our first full-scale prototype. This version emphasizes both functionality and sustainability by using repurposed materials, including a metal structure from decommissioned antenna ladders and large wooden cable reels that were no longer in use.
- The design also incorporates side spaces for banners, allowing for educational content or partnerships to be displayed directly on the bin. As development progresses, we are actively integrating the physical structure with electronic components, bringing Binnsie closer to a fully operational system designed for real-world impact.

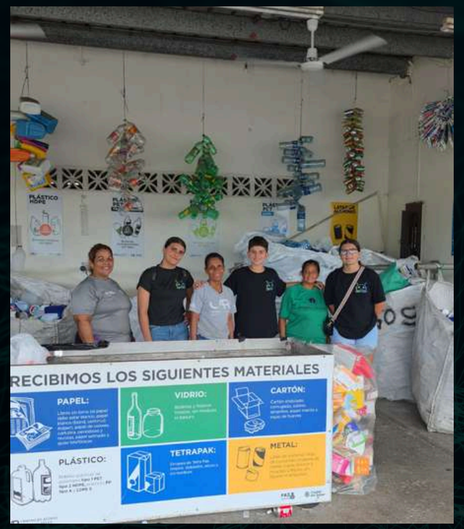
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BINNSIE

Supporting document #2

Development Process















- Beyond UX research, we have conducted field visits to better understand the real-world dynamics of recycling and to begin building potential partnerships. These visits allowed us to explore operational costs, identify practical pathways to achieve our goals, and validate the problem beyond online research.
- We visited multiple recycling centers and waste collection sites, including Fundación Bliss Panamá, where materials are not only collected but also transformed into products with commercial value. During our visit, we received a guided tour, gained insights into the current state of recycling in Panama, and discussed potential collaboration opportunities aligned with a shared mission.
- Additional visits, such as to FAS Panamá, further strengthened our understanding of material flow and reinforced the importance of efficient sorting at the source.